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EXERCISE 4 - JOINS SQL

QUESTION 1

```sql
-- Q1: users (A) + subscriptions (B)
-- Expected: user_id, user_name, subscription_id, start_date
SELECT
  A.user_id,
  A.user_name,
  B.subscription_id,
  B.start_date
FROM users AS A
INNER JOIN subscriptions AS B ON A.user_id = B.user_id;
```

Add parameter

Table

	user_id	user_name	subscription_id	start_date
1	1	Nomvula	503	2026-03-10
2	1	Nomvula	501	2026-01-15
3	2	David	502	2026-02-01
4	3	Anele	505	2026-04-05

QUESTION 2

```sql
-- Q2: subscriptions (A) + plans (B)
SELECT
  A.subscription_id,
  A.user_id,
  B.plan_name,
  B.monthly_price
FROM subscriptions AS A
INNER JOIN plans AS B ON A.plan_id = B.plan_id;
```

Add parameter

Table

	subscription_id	user_id	plan_name	monthly_price
1	501	1	Basic	79
2	504	6	Standard	129
3	502	2	Standard	129
4	503	1	Premium	199

QUESTION 3

```sql
-- Q3: viewing_sessions (A) + shows (B)
SELECT
  A.session_id,
  A.user_id,
  B.show_title,
  B.genre,
  A.watch_minutes
FROM viewing_sessions AS A
INNER JOIN shows AS B ON A.show_id = B.show_id;
```

Add parameter

Table

	session_id	user_id	show_title	genre	watch_minutes
1	904	7	Comedy Hour	Comedy	20
2	901	1	Comedy Hour	Comedy	45
3	903	1	Crime Time	Drama	60
4	902	2	Tech Tales	Documentary	30

QUESTION 4

```sql
-- Q4: users (A) + viewing_sessions (B)  -- Note: prompt says users · viewing_sessions
SELECT
  A.user_name,
  A.country,
  B.session_id,
  B.show_id,
  B.watch_minutes
FROM users AS A
INNER JOIN viewing_sessions AS B ON A.user_id = B.user_id;
```

Add parameter

Table

	user_name	country	session_id	show_id	watch_minutes
1	Nomvula	Johannesburg	903	702	60
2	Nomvula	Johannesburg	901	701	45
3	David	Cape Town	902	703	30
4	Anele	Durban	905	705	90

QUESTION 5

```sql
-- Q5: users (A) + subscriptions (B) + plans (C) - 3 tables with A,B,C
SELECT
  A.user_name,
  A.country,
  C.plan_name,
  C.monthly_price,
  B.start_date
FROM users AS A
INNER JOIN subscriptions AS B ON A.user_id = B.user_id
INNER JOIN plans AS C ON B.plan_id = C.plan_id;
```

Add parameter

Table

	user_name	country	plan_name	monthly_price	start_date
1	Nomvula	Johannesburg	Basic	79	2026-01-15
2	David	Cape Town	Standard	129	2026-02-01
3	Nomvula	Johannesburg	Premium	199	2026-03-10

QUESTION 6

```sql
-- Q6: users (A) LEFT subscriptions (B)
SELECT
  A.user_id,
  A.user_name,
  B.subscription_id,
  B.start_date
FROM users AS A
LEFT JOIN subscriptions AS B ON A.user_id = B.user_id;
```

Add parameter

Table

	user_id	user_name	subscription_id	start_date
1	1	Nomvula	503	2026-03-10
2	1	Nomvula	501	2026-01-15
3	2	David	502	2026-02-01
4	3	Anele	505	2026-04-05
5	4	Kabelo	null	null

QUESTION 7

```sql
-- Q7: plans (A) LEFT subscriptions (B)
SELECT
  A.plan_id,
  A.plan_name,
  B.subscription_id,
  B.user_id
FROM plans AS A
LEFT JOIN subscriptions AS B ON A.plan id = B.plan id;
```

Add parameter

Table

	plan_id	plan_name	subscription_id	user_id
1	10	Basic	501	1
2	11	Standard	504	6
3	11	Standard	502	2
4	12	Premium	503	1
5	13	Family	505	3
6	14	Mobile	null	null

QUESTION 8

```sql
-- Q8: shows (A) LEFT viewing_sessions (B)
SELECT
  A.show_id,
  A.show_title,
  B.session_id,
  B.watch_minutes
FROM shows AS A
LEFT JOIN viewing_sessions AS B ON A.show_id = B.show_id;
```

Add parameter

Table

	show_id	show_title	session_id	watch_minutes
1	701	Comedy Hour	904	20
2	701	Comedy Hour	901	45
3	702	Crime Time	903	60
4	703	Tech Tales	902	30
5	704	Cooking Lab	null	null

QUESTION 9

```sql
-- Q9: viewing_sessions (A) LEFT users (B)
SELECT
  A.session_id,
  A.show_id,
  A.watch_minutes,
  A.user_id,
  B.user_name
FROM viewing_sessions AS A
LEFT JOIN users AS B ON A.user_id = B.user_id;
```

Add parameter

Table

	session_id	show_id	watch_minutes	user_id	user_name
1	901	701	45	1	Nomvula
2	902	703	30	2	David
3	903	702	60	1	Nomvula
4	904	701	20	7	null
5	905	705	90	3	Anele

QUESTION 10

```
148  -- Q10: users (A) LEFT subscriptions (B) LEFT plans (C)
149  SELECT
150      A.user_name,
151      A.country,
152      C.plan_name,
153      C.monthly_price
154  FROM users AS A
155  LEFT JOIN subscriptions AS B ON A.user_id = B.user_id
156  LEFT JOIN plans AS C ON B.plan_id = C.plan_id;
```

Add parameter

Table

	user_name	country	plan_name	monthly_price
1	Nomvula	Johannesburg	Premium	199
2	Nomvula	Johannesburg	Basic	79
3	David	Cape Town	Standard	129
4	Anele	Durban	Family	249
5	Kabelo	Pretoria	null	null

QUESTION 11

```
162  -- Q11: users (A) FULL OUTER subscriptions (B)
163  SELECT
164      A.user_id,
165      A.user_name,
166      B.subscription_id,
167      B.start_date
168  FROM users AS A
169  FULL OUTER JOIN subscriptions AS B ON A.user_id = B.user_id;
```

Add parameter

Table

	user_id	user_name	subscription_id	start_date
1	1	Nomvula	503	2026-03-10
2	2	David	502	2026-02-01
3	3	Anele	505	2026-04-05
4	4	Kabelo	null	null

QUSTION 12

```sql
-- Q12: plans (A) FULL OUTER subscriptions (B)
SELECT
  A.plan_id,
  A.plan_name,
  B.subscription_id,
  B.user_id
FROM plans AS A
FULL OUTER JOIN subscriptions AS B ON A.plan_id = B.plan_id;
```

Add parameter

Table

	plan_id	plan_name	subscription_id	user_id
1	10	Basic	501	1
2	11	Standard	504	6
3	12	Premium	503	1
4	13	Family	505	3
5	14	Mobile	null	null

QUESTION 13

```sql
-- Q13: shows (A) FULL OUTER viewing_sessions (B)
SELECT
  A.show_id,
  A.show_title,
  B.session_id,
  B.watch_minutes
FROM shows AS A
FULL OUTER JOIN viewing_sessions AS B ON A.show_id = B.show_id;
```

Add parameter

Table

	show_id	show_title	session_id	watch_minutes
1	701	Comedy Hour	904	20
2	702	Crime Time	903	60
3	703	Tech Tales	902	30
4	704	Cooking Lab	null	null
5	706	Wild Earth	null	null

QUESTION 14

```sql
-- Q14: users (A) FULL OUTER viewing_sessions (B)
SELECT
  A.user_id,
  A.user_name,
  B.session_id,
  B.show_id,
  B.watch_minutes
FROM users AS A
FULL OUTER JOIN viewing_sessions AS B ON A.user_id = B.user_id;
```

Add parameter

Table

	user_id	user_name	session_id	show_id	watch_minutes
1	1	Nomvula	901	701	45
2	2	David	902	703	30
3	1	Nomvula	903	702	60
4	null	null	904	701	20

QUESTION 15

```sql
-- Q15: users (A) FULL OUTER subscriptions (B) FULL OUTER plans (C)
SELECT A.user_id,
  A.user_name,
  B.subscription_id,
  C.plan_id,
  C.plan_name
FROM users AS A
FULL OUTER JOIN subscriptions AS B ON A.user_id = B.user_id
FULL OUTER JOIN plans AS C ON B.plan_id = C.plan_id;
```

Add parameter

Table

	user_id	user_name	subscription_id	plan_id	plan_name
1	1	Nomvula	503	12	Premium
2	1	Nomvula	501	10	Basic
3	3	Anele	505	13	Family
4	4	Kabelo	null	null	null

BONUS ANSWERS

1.

```sql
-- BONUS 01: users without subscriptions
SELECT A.user_id, A.user_name
FROM users AS A
LEFT JOIN subscriptions AS B ON A.user_id = B.user_id
WHERE B.subscription_id IS NULL;
```

Add parameter

Table

	user_id	user_name
1	4	Kabelo
2	5	Lerato

2.

```sql
-- BONUS 02: subscriptions with no valid user
SELECT A.*
FROM subscriptions AS A
LEFT JOIN users AS B ON A.user_id = B.user_id
WHERE B.user_id IS NULL;
```

Add parameter

Table

	subscription_id	user_id	plan_id	start...
1	504	6	11	2026-03-20

3.

```sql
-- BONUS 03: shows never watched
SELECT A.show_id, A.show_title
FROM shows AS A
LEFT JOIN viewing_sessions AS B ON A.show_id = B.show_id
WHERE B.session_id IS NULL;
```

Add parameter

Table

	show_id	show_title
1	704	Cooking Lab
2	706	Wild Earth

4.

```sql
-- BONUS 04: sessions with invalid show
SELECT A.*
FROM viewing_sessions AS A
LEFT JOIN shows AS B ON A.show_id = B.show_id
WHERE B.show_id IS NULL;
```

Add parameter

Table

	session_id	user_id	show_id	watch_minutes
1	905	3	705	90

5.

```sql
-- BONUS 05: plans with no subscribers
SELECT A.plan_id, A.plan_name
FROM plans AS A
LEFT JOIN subscriptions AS B ON A.plan_id = B.plan_id
WHERE B.subscription_id IS NULL;
```

Add parameter

Table

	plan_id	plan_name
1	14	Mobile